Course 6032B · Semester 6 · Program Elective · Revision 2021

Energy Conservation & Management

A professional engineering handbook on renewable energy sources, energy management, energy efficiency, BEE star labelling, energy conservation in buildings, HVAC and lighting savings, electrical energy audit methodology and Sankey diagrams.

What this subject is

This course explains why energy conservation is necessary and how an electrical engineer can reduce energy loss in real systems. It begins with renewable energy sources, then moves into energy scenario, energy management techniques, energy efficiency, building energy conservation, HVAC, lighting, energy audit instruments, audit stages and energy flow representation.

Malayalam: Energy conservation □□□□□□ output □□□□□□□□□□ □□□□□□□. □□□ useful work □□□□□ energy □□□□□□□□□□ □□□□□□□ □□□□□□□□. Factory, building, panel, motor, lighting, HVAC □□□□□□□□□□□□□□ losses □□□□□□□□□ □□□□□□□□□□□□□□□□ □ subject-□□□□ practical value.

Official syllabus information

Item	Value
Program	Diploma in Electrical and Electronics Engineering
Course code	6032B
Course title	Energy Conservation & Management
Semester	6
Credits	4
Category	Program Elective
Periods	4 per week (L:3 T:1 P:0); 60 per semester
CO-PO Mapping	CO1-CO4 moderately mapped to PO1

Course objectives

- Address energy crisis and related issues.
- Identify the need of energy conservation.
- Observe standard practices based on Energy Conservation Act.
- Familiarize energy auditing and apply engineering practices to reduce energy loss.

Prerequisites

- Power and energy from Applied Physics-I, Semester 1.
- Power and energy from Fundamentals of Electrical & Electronics Engineering, Semester 2.

Course outcomes

- CO1: Explain renewable energy sources and importance.
- CO2: Summarize energy management techniques.
- CO3: Explain energy conservation practices in buildings.
- CO4: Summarize energy audit in electrical systems.

Redesigned syllabus roadmap

The official module outcomes converted into a practical energy engineering study route.

Module 1 · CO1 · 15 Hours

Renewable Energy Sources and Importance

Energy classification, renewable energy need, solar, wind, ocean, geothermal, biogas and biomass energy conversion.

Classification of energy sources

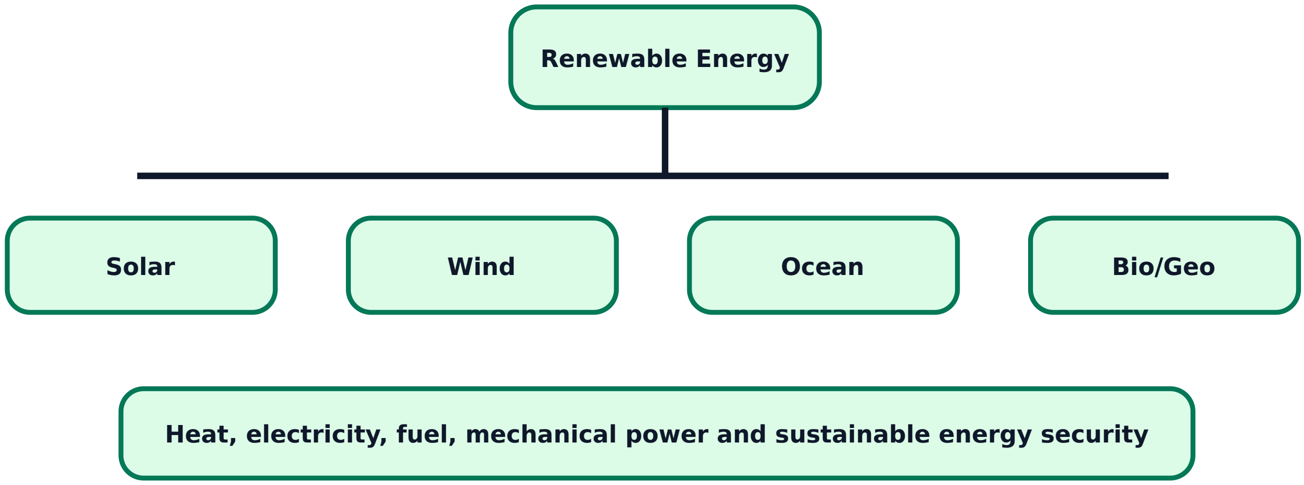
Energy sources are commonly classified as conventional and non-conventional, renewable and non-renewable, commercial and non-commercial. Conventional sources such as coal, petroleum and natural gas are widely used but cause depletion and environmental impact. Renewable sources such as solar, wind, ocean, geothermal and biomass are naturally replenished.

Malayalam: Coal, oil □□□□□□□ conventional fuels limited □□□. Solar, wind □□□□□□□ renewable sources □□□□□□□ □□□□□□□□□□□□□□□□. Energy crisis □□□□□□□□□□ conservation □□□□ renewable adoption □□□□.

Category	Examples	Engineering concern
Conventional	Coal, oil, natural gas	Reserve depletion, pollution
Renewable	Solar, wind, ocean, geothermal, biomass	Intermittency, site dependency

Commercial	Electricity, LPG, petrol	Billing and efficiency matter
Non-commercial	Firewood, agricultural residues	Low efficiency and sustainability issues

Renewable energy conversion map



Solar energy

Solar energy is radiation from the sun used for heat and electricity. Solar PV cells convert light directly to DC electricity. Solar thermal systems use heat for water heating, drying and process heating. Important terms are solar radiation, solar irradiation and solar insolation.

> **Exam tip:** For solar questions, write basic principle, advantages, disadvantages and applications separately.

Wind and ocean energy

Wind energy uses kinetic energy of moving air to rotate turbine blades and generate electricity. A typical wind energy system includes rotor, blades, shaft, gearbox or direct drive, generator, tower, controller and power conditioning unit. Ocean energy includes tidal energy and wave energy. It is site-specific but predictable in some locations.

Geothermal, biogas and biomass

Geothermal energy uses heat from inside the earth. Biogas is produced by anaerobic digestion of organic matter and mainly contains methane and carbon dioxide. Fixed dome and floating drum biogas plants are common. Biomass includes agricultural residues, wood, animal waste and organic municipal waste.

Expected questions

1. Classify energy sources with examples.

2. Explain need, advantages and disadvantages of renewable energy.
3. Explain solar PV cell and solar energy terms.
4. Draw components of typical wind energy system.
5. Explain fixed dome and floating drum biogas plants.

Module 2 · CO2 · 14 Hours

Energy Scenario and Energy Management Techniques

Energy scenario, energy management definition, objectives, functions, techniques, efficiency, conservation and BEE star labelling.

Energy scenario

The syllabus expects review of global energy scenario, coal/oil/natural gas reserves, consumption in industrial, agriculture and residential sectors in India, long-term energy scenario and environmental impacts of conventional energy use. The key point is simple: demand keeps increasing, fossil reserves are limited, and environmental pressure forces conservation and renewable integration.

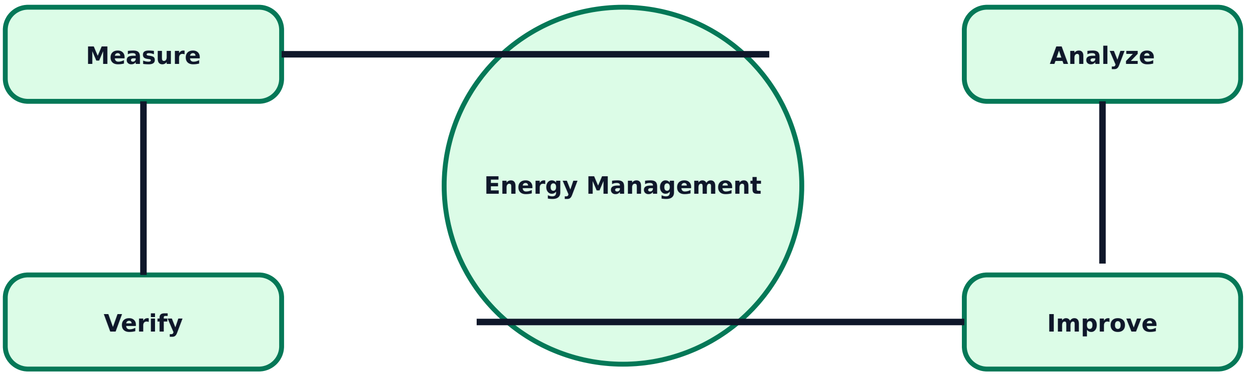
Malayalam: Energy demand □□□□□□□□□ fuel cost, pollution, power shortage □□□□□□□□□ □□□□□□. □□□□□□□ efficiency improvement, conservation, renewable usage □□□□□□□ industrial decision □□□□□.

Energy management

Energy management is the planned monitoring, control and improvement of energy use without reducing useful output. Objectives include reducing cost, reducing losses, improving efficiency, improving reliability and reducing environmental impact.

- Measure energy consumption.
- Identify high-consumption areas.
- Evaluate losses and inefficiencies.
- Implement technical and operational improvements.
- Verify savings and continue monitoring.

Energy management cycle



Energy management techniques

Techniques include self-knowledge and awareness, reengineering and evaluation, technology upgrades, monitoring, benchmarking and employee participation. In an electrical department this means load study, avoiding idle running, power factor correction, efficient motors, efficient lighting, right cable sizing and maintenance-based loss reduction.

Energy efficiency and star labelling

Energy efficiency means same useful output with less energy input. Energy conversion efficiency is useful output divided by input. Energy conservation is reducing wasteful use of energy. BEE star labelling helps consumers compare appliance energy performance. Comparative labels rank products; endorsement labels certify that a product meets an efficiency requirement.

Expected questions

1. Explain current energy scenario in India.
2. Define energy management and list objectives.
3. Explain energy management techniques.
4. Explain energy efficiency, energy conversion efficiency and energy conservation.
5. Explain BEE star labelling, types of labels and benefits.

Module 3 · CO3 · 14 Hours

Energy Conservation in Buildings

Energy conservation principles, domestic/commercial practices, HVAC opportunities, thermal insulation and lighting energy savings.

Principles of energy conservation

Energy conservation means reducing unnecessary energy use while maintaining required comfort, safety and productivity. Principles include avoiding wastage, improving efficiency, recovering losses, using correct equipment, operating at proper loading and maintaining equipment.

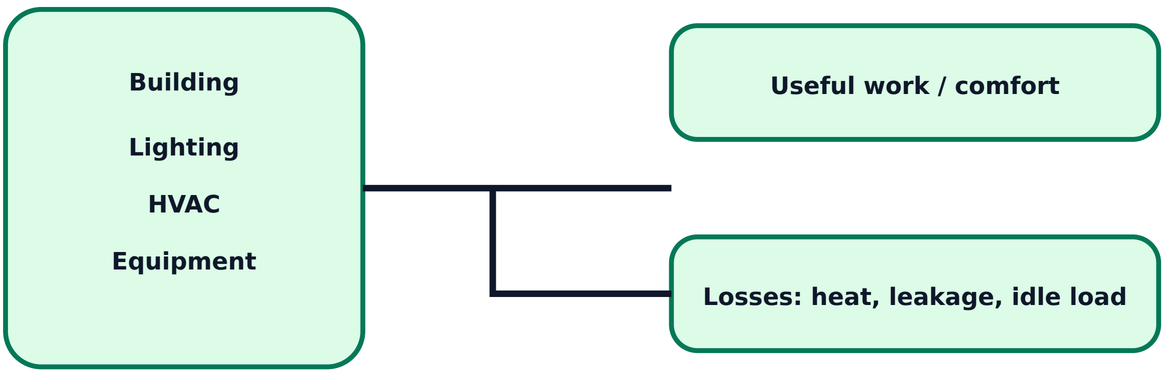
Malayalam: Building-□ conservation □□□□□□□□□□□ light off □□□□□□□ □□□□□□ □□□□. HVAC load, insulation, daylight, efficient lamps, control sensors, maintenance □□□□□□ □□□□□□ check □□□□□□□.

Building conservation activities

Domestic and commercial sectors can save energy through efficient appliances, proper scheduling, daylight use, power factor correction where applicable, occupancy sensors, efficient pumps, good insulation, energy metering and regular maintenance. Commercial buildings should monitor load profile and peak demand.

Area	Action	Benefit
Lighting	LED, daylight, occupancy sensor	Lower kWh and heat load
HVAC	Correct thermostat, filter cleaning, insulation	Lower compressor/fan energy
Motors/Pumps	Correct sizing, VFD where useful	Reduced running cost
Envelope	Roof/wall insulation and shading	Lower cooling load

Building energy flow



HVAC conservation opportunities

HVAC system components include compressor, condenser, evaporator, expansion device, fans, ducts, filters and controls. Opportunities include proper temperature setting, filter cleaning, duct leakage control, thermal insulation, efficient chillers, efficient fans, VFD for variable air volume systems and reducing unwanted heat gain.

Lighting conservation

Lighting savings can be achieved by using LED lamps, efficient luminaires, correct lux level, daylight harvesting, occupancy sensors, timers, cleaning lamps/luminaires and using task lighting instead of over-lighting the whole area.

Expected questions

1. Define energy conservation and explain its importance.
2. List energy saving tips for domestic and commercial buildings.
3. Explain HVAC energy conservation opportunities.
4. Explain thermal insulation and its relation to efficiency.
5. Explain lighting energy conservation techniques.

Module 4 · CO4 · 15 Hours

Energy Audit of Electrical Systems

Energy audit definition, objectives, BEE regulations, audit types, instruments, stages, ten-step methodology, report format and Sankey diagram.

Energy audit meaning and objectives

Energy audit is a systematic inspection, measurement and analysis of energy use to identify saving opportunities. Objectives are to find where energy is consumed, identify avoidable losses, recommend improvements, estimate savings and payback, and prepare a practical action plan.

Malayalam: Energy audit guess work □□□□. Measurement, data sheet, bill analysis, instrument reading, observation □□□□□□□□□□ □□□□□□□□□□□□□ loss □□□□□□□□□ correction plan □□□□□□□□□□□.

Types of energy audit

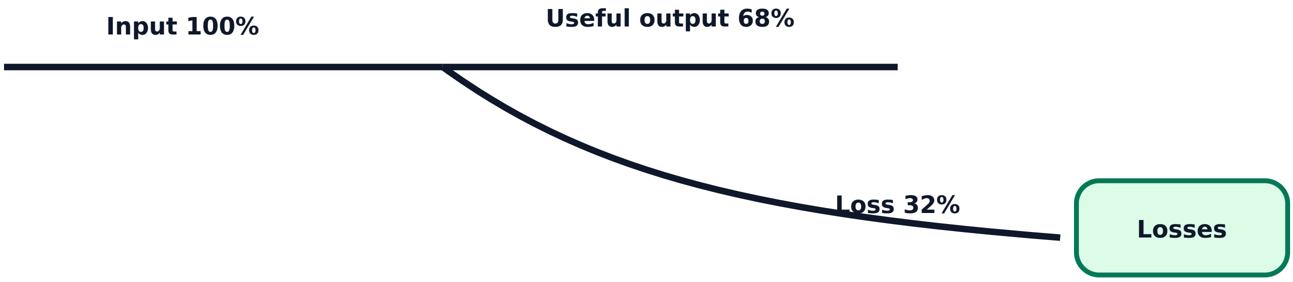
- **Preliminary audit:** quick walk-through survey and initial saving opportunities.
- **Targeted audit:** focused study of a selected system such as lighting, motor or HVAC.
- **Detailed audit:** complete measurement, analysis, techno-economic recommendation and report.

Audit instruments

Instrument	Purpose
Energy meter	kWh consumption measurement
Power analyser	Voltage, current, power, PF and harmonics
Clamp meter	Current measurement without opening circuit

Lux meter	Lighting level measurement
Tachometer	Motor/fan speed
Thermometer / thermal imager	Temperature and hot spot detection
Data logger	Long-term load profile recording

Sankey diagram concept



Audit methodology

Typical stages include formation of audit team, collection of utility bills, preparation of data sheets, walk-through survey, measurement, load analysis, loss identification, recommendation, savings/payback calculation, report preparation and post-implementation verification.

> **Practical note:** Never recommend capacitor bank, VFD, motor replacement or lighting retrofit without measuring load, operating hours and existing performance.

Expected questions

1. Define energy audit and explain objectives.
2. List instruments used for energy audit and their purpose.
3. Explain preliminary, targeted and detailed audit.
4. Explain methodology for conducting energy audit.
5. Explain Sankey diagram and its role in energy analysis.

Formula / Rule Bank

Energy conservation calculations and audit rules used in exam answers and field work.

Energy consumption

Energy (kWh) = Power (kW) x Time (h)

Cost of energy

Cost = Energy consumed (kWh) x tariff per kWh

Energy saving

Saving = Old consumption - New consumption

Annual saving

Annual saving = Daily saving x operating days per year

Simple payback

Payback = Investment / Annual cost saving

Efficiency

Efficiency = Useful output / Input x 100%

Lighting retrofit rule

Power saved = number of lamps x (old wattage - new wattage)

Audit rule

Measure first, analyze second, recommend third, verify after implementation.

Solved Examples / Problem Lab

Step-by-step examples for conservation and audit calculation.

Example 1 - Lighting saving

Given: 50 lamps of 40 W replaced by 18 W LED. Usage = 8 h/day.

Power saved = 50 x (40-18) W = 1100 W = 1.1 kW.

Daily saving = 1.1 x 8 = 8.8 kWh.

Monthly saving for 30 days = 264 kWh.

Example 2 - Cost saving

Energy saving = 264 kWh/month. Tariff = Rs.7/kWh.

Monthly cost saving = 264 x 7.

Cost saving = Rs.1848/month.

Example 3 - Payback

Investment = Rs.18000. Annual saving = Rs.9000.

Payback = 18000 / 9000.

Simple payback = 2 years.

Example 4 - Motor idle loss

A 2 kW idle load runs unnecessarily for 3 h/day.

Daily wasted energy = 2 x 3 = 6 kWh.

At Rs.8/kWh, daily waste = Rs.48.

Example 5 - Efficiency

Input = 10 kW, useful output = 8 kW.

Efficiency = 8/10 x 100.

Efficiency = 80%.

Practice Section and Expected Questions

Module-wise exam preparation for Energy Conservation & Management.

One-mark questions

1. Define energy conservation.
2. What is energy audit?
3. What is BEE?
4. What is Sankey diagram?
5. Name two audit instruments.

Short-answer questions

1. Classify energy sources with examples.
2. List components of wind energy system.
3. Explain energy management objectives.
4. Explain BEE star labelling.
5. List HVAC conservation opportunities.

Descriptive questions

1. Explain renewable energy sources and their importance.
2. Explain energy management techniques.

3. Explain energy conservation in domestic and commercial buildings.
4. Explain methodology for conducting energy audit.

MCQ practice

1. A lux meter measures: A) current B) illumination C) energy D) speed

2. Simple payback is: A) Investment/annual saving B) annual saving/investment C) voltage/current D) kWh/tariff

3. Sankey diagram mainly shows: A) wiring only B) energy flow and losses C) star rating D) tariff slab

Fresh Model Question Paper

Original model paper based on syllabus coverage; not copied from official question papers.

Course 6032B - Energy Conservation & Management

Time: 3 Hours **Maximum Marks:** 100

Part A - Short Answer

1. Define renewable energy.
2. What is energy efficiency?
3. What is BEE star label?
4. Name any two HVAC components.
5. What is preliminary energy audit?

Part B - Medium Answer

1. Classify energy sources with examples.
2. Explain solar, wind and ocean energy conversion systems.
3. Explain energy management techniques.
4. Explain energy conservation opportunities in lighting systems.
5. List instruments used in energy audit and write their purpose.

Part C - Long Answer / Problems

1. Explain biogas plant construction and working with neat sketch.
2. Explain energy scenario, environmental impacts and solutions.
3. Explain energy conservation activities in buildings including HVAC and lighting.
4. A building replaces 80 lamps of 40 W with 18 W LED lamps. Usage is 10 h/day and tariff is Rs.8/kWh. Calculate daily energy saving and daily cost saving.
5. Explain detailed energy audit methodology and Sankey diagram.

Complete Model Answers

Answer guide with formulas, diagram points and common mistakes.

Renewable energy definition

Renewable energy is energy obtained from sources that are naturally replenished, such as solar, wind, ocean, geothermal and biomass.

Lighting numerical answer

Power saved = 80 x (40-18) W = 1760 W = 1.76 kW. Daily saving = 1.76 x 10 = 17.6 kWh. Cost saving = 17.6 x 8 = Rs.140.8 per day.

Daily energy saving = 17.6 kWh; daily cost saving = Rs.140.8.

Energy audit methodology points

Form audit team, collect bills, prepare data sheets, perform walk-through survey, measure loads, analyze data, identify loss areas, suggest measures, calculate saving/payback, prepare report and verify implementation.

Sankey answer points

A Sankey diagram shows energy input, useful output and losses using arrows whose width is proportional to energy quantity. It helps identify major loss paths.

Common mistakes

Do not write generic awareness points only. In engineering answers, mention measurement, load, operating hours, power rating, efficiency, cost saving and payback.

References from syllabus

BEE guide books, Dr. Sanjeev Singh, O.P. Gupta, G.D. Rai, P.D. Henderson, W.C. Turner, K.V. Sharma, V.K. Mehta, BEE India, Indian Power Sector, Kerala Energy and Ministry of Power.